A MAP OF MODULES WHOSE INDUCED MAP ON FIRST TOR VANISHES FOR ALL TEST MODULES FACTORS THROUGH A FREE MODULE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter source sources/stacks-local-cohomology/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. Selection provenance: this card was opened under the file-based control contract (CONTROL.md), with the mechanical pre-card gate run on its statement and proof interval before the card was written.

Allowed standard prerequisites (editorial).

- The first Tor functor and the presentation of a finitely presented module as the cokernel of a map of finite free modules. Used at the start of the counted proof (source lines 4307-4321). Supplies the two short exact sequences that the proof compares.
- Universal exactness of a short exact sequence and the criterion that a universally exact sequence of modules with the middle term as displayed splits. Invoked inside the counted proof at source lines 4331-4335 and attributed to a lemma of another chapter (a cross-chapter pointer). Converts the vanishing of the boundary map into a splitting, which yields the required factorisation.

Source reference. Original reference algebra-lemma-universally-exact-split: sources/stacks-local-cohomology/pinned-original-source.tex, source lines 4333–4333. Navigation pointer only: the counted text refers to the Stacks item algebra-lemma-universally-exact-split, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Lemma 0.1. *Let R be a ring. Let $M \to M'$ be a map of R-modules with M of finite presentation such that $Tor_1^R(M, N) \to Tor_1^R(M', N)$ is zero for all R-modules N. Then $M \to M'$ factors through a free R-module.*

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Proof. We may choose a map of short exact sequences

$$\begin{array}{ccccccccc} 0 & \longrightarrow & K & \longrightarrow & R^{\oplus r} & \longrightarrow & M & \longrightarrow & 0 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & K' & \longrightarrow & \bigoplus_{i \in I} R & \longrightarrow & M' & \longrightarrow & 0 \end{array}$$

whose right vertical arrow is the given map. We can factor this map through the short exact sequence

$$0 \to K' \to E \to M \to 0 \tag{0.1.1}$$

which is the pushout of the first short exact sequence by $K \to K'$. By a diagram chase we see that the assumption in the lemma implies that the boundary map $\mathrm{Tor}_1^R(M, N) \to K' \otimes_R N$ induced by (0.1.1) is zero, i.e., the sequence (0.1.1) is universally exact. This implies by Algebra, Lemma algebra-lemma-universally-exact-split that (0.1.1) is split (this is where we use that M is of finite presentation). Hence the map $M \to M'$ factors through $\bigoplus_{i \in I} R$ and we win. □